

Osama Ansar

Senior Software Engineer — AI/RAG Systems & Full-Stack Platforms

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SUMMARY

Senior Software Engineer with 7+ years shipping scalable, high-performance web applications across SaaS, healthcare technology, logistics, streaming, and enterprise platforms — now specializing in TypeScript-based RAG systems, agentic pipelines, and LLM evaluation infrastructure. Delivered production AI features including a Model Context Protocol server and LLM-powered document extraction, alongside a self-directed portfolio of from-scratch RAG, vector-search, and agent-evaluation systems and five accepted open-source contributions across the Weaviate, Qdrant, Turso, and Google Agent2Agent TypeScript SDKs.

TECHNICAL SKILLS

AI / RAG: Retrieval-Augmented Generation, Vercel AI SDK, LangChain.js, LangGraph.js, BM25, HNSW & vector search, LLM evaluation (LLM-as-judge), prompt engineering, Model Context Protocol (MCP), agentic tool-calling, OAuth 2.1 / PKCE

Languages: TypeScript, JavaScript, Python, SQL

Frontend: React, Next.js, Angular, Vue, RxJS, Tailwind CSS, SCSS

Backend: Node.js, Koa.js, Hono.js, Express.js, ASP.NET Core Web API, FastAPI

Data: PostgreSQL, pgvector, Sequelize, Drizzle ORM, Entity Framework Core, MongoDB

Architecture: Microservices, microfrontends, REST APIs, real-time systems, distributed systems, SaaS

Cloud / DevOps: AWS, GCP, Cloudflare Workers, Docker, Kubernetes, Terraform, GitHub Actions, GitLab CI/CD, Jenkins

Payments / Streaming: Stripe, Montonio, Mollie, Mux, AWS IVS, Dolby, DRM

PROFESSIONAL EXPERIENCE

MenuLogic K12 — Senior Software Engineer

Nov 2025 – Present | United States

- Built the product's Model Context Protocol (MCP) server, exposing recipe, ingredient, and district tools to AI agents (Claude in Chat/Cowork) via a custom OAuth 2.1 + PKCE authorization-code flow and tenant-scoped tool authorization that mirrors the REST API's own role system.
- Build and maintain full-stack features using Angular, standalone components, RxJS, and Kendo UI on the frontend with ASP.NET Core Web API and Entity Framework Core on the backend.
- Develop USDA-compliance reporting features with Telerik Reporting, including report data pipelines, EF Core queries, REST data-source endpoints, and parameter-driven report definitions.
- Implement cross-application SSO between a legacy MVC shell and an Angular SPA using JWT token handoff via iframe messaging.

Truckpedia — Software Engineer

Jan 2024 – Oct 2025 | United States

- Led frontend development using React and Vue across a logistics platform.
- Integrated AI-driven features including OpenAI/LLM-powered prompt engineering and knowledge-base chat for intelligent, context-aware interactions.
- Optimized scalability and user performance and implemented real-time updates across multiple devices.
- Collaborated across teams, led feature implementation, mentored junior engineers, and maintained code quality.

Omnivex Corporation — Software Engineer

May 2022 – Dec 2023 | Canada

- Developed custom UI components for the Ink platform using Angular, React, and Vue.
- Built real-time user experiences with SignalR and improved component reusability while reducing code duplication.
- Led bug triage for ASP.NET and Angular frontend issues.

Contrive Solutions — Senior JavaScript Developer

Jan 2020 – Jun 2022 | Pakistan

- Built Node.js automation for S4 Salesforce bug reporting.
- Developed the GoOrder food-delivery application using Vue/Vuex and the Penmi calendar/CRUD features using Angular.
- Built reusable UI components across seven products and shared Angular microfrontend component libraries for Finlex.

Webicosoft — JavaScript Developer

May 2019 – Nov 2019 | Pakistan

- Developed full-stack features for the Blunk job-matching platform using React, Redux, Semantic UI, Node.js, and MongoDB.

AI / RAG ENGINEERING — INDEPENDENT PROJECTS

MELAI Engineering Lab

github.com/OsamaAnsar/MELAI-Engineering-lab

An LLM evaluation and agent workbench: hand-written BM25/vector/hybrid (RRF) retrieval compared side by side, an LLM-as-judge scoring pipeline (recall@k, precision@k, MRR, nDCG, hit-rate), an agent tool-calling loop, and a CLI that gates CI on a regression eval. Provider SDKs sit behind one interface; every run persists the fully-resolved request so it replays exactly. pnpm/Turborepo monorepo, Fastify + SSE, Drizzle + pgvector.

Refract

github.com/OsamaAnsar/refract-rag

A hybrid RAG system that shows its work: from-scratch BM25 (lexical) and HNSW (vector) retrieval run in parallel and get fused with Reciprocal Rank Fusion, with multi-hop agentic decomposition for cross-topic questions. The UI visualizes every retrieval score, fusion rank, and sub-question live as it runs.

rag-eval-harness

github.com/OsamaAnsar/rag-eval-harness

LLM-judge evaluation harness for RAG pipelines — faithfulness, answer relevance, context precision, and context recall — with a CI regression gate that catches planted hallucinations in a benchmark dataset. Pluggable Judge interface (real OpenAI judge plus a dependency-free FakeJudge for CI).

agentic-rag

github.com/OsamaAnsar/agentic-rag

Multi-hop agentic RAG pipeline: decomposes complex questions into sub-questions, retrieves evidence with a from-scratch BM25 implementation, checks answer sufficiency, and synthesizes a cited final answer.

hnsw-vector-index

github.com/OsamaAnsar/hnsw-vector-index

HNSW (Hierarchical Navigable Small World) approximate-nearest-neighbor index implemented from scratch, no ANN library — recall@k climbs from 66% to 100% against a brute-force baseline as efSearch increases.

generative-ui-chat & doc-chat-citations

Two Vercel AI SDK tool-calling chat assistants: one streams live, interactive React components (charts, tables, booking cards) instead of plain text; the other grounds every answer in from-scratch BM25 retrieval with clickable, source-linked citations.

OPEN SOURCE

Vector-DB & Agent SDKs

github.com/OsamaAnsar

Five accepted pull requests across four official TypeScript SDKs: a maintainer-approved generative-module config builder and a silent data-loss fix in **Weaviate**'s client; a dropped reverse-proxy path prefix in **Qdrant**'s URL parsing; a UTF-8 vs UTF-16 byte-limit mismatch in **Turso** (libSQL)'s SQL cache; and pagination silently ignored by **Google Agent2Agent**'s REST transport. Each PR mirrors the target repo's existing patterns and ships with regression tests proving the bug and the fix.

SELECTED CLIENT PROJECTS

GoStork

AI-powered fertility services marketplace

Built an AI-powered PDF extraction pipeline using GPT-4 to transform unstructured agency documents into standardized, searchable donor and provider profiles — structured data mapping, cost/compensation analysis, image analysis, and summary generation. Built real-time messaging infrastructure; integrated PostgreSQL/Sequelize, HubSpot, Stripe, Google Calendar, Outlook, Twilio, and SendGrid on Docker/Kubernetes/GCP.

Streamflow / LiveArena

Pay-per-view live streaming platform

Built DRM-protected streaming and an event-based pricing engine for an enterprise PPV platform designed for up to 500,000 concurrent users. Implemented real-time payment celebrations and dynamic watermarking via Pusher, resumable S3 uploads with TUS/Uppy, and seamless player switching during buffering. Stack: Next.js/React, Hono, Cloudflare Workers, PostgreSQL, AWS Lambda/SQS, Stripe, Mux/AWS IVS.

EDUCATION

BS Software Engineering

University of Gujrat | 2015–2019 | CGPA 3.7/4.0

Artificial Intelligence | Machine Learning | Data Structures & Algorithms | Database Management Systems | Software Engineering

AWARDS

- Best Performance Award — Contrive Solutions
- 1st Place — Speed Programming Competition
- Merit Scholarship Recipient
- UOG Topper — 1st Position
- Selected for Business Incubation Program